

Deep Quant Finance (Orientation)

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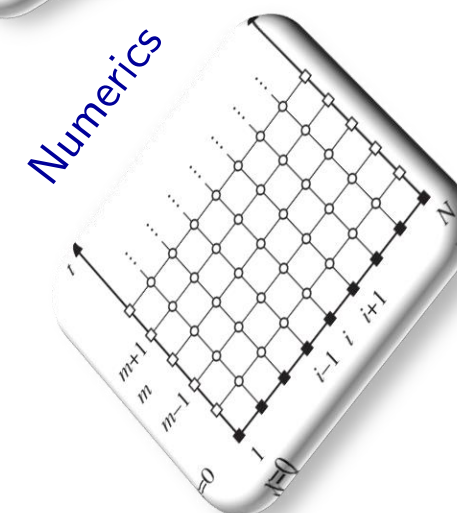
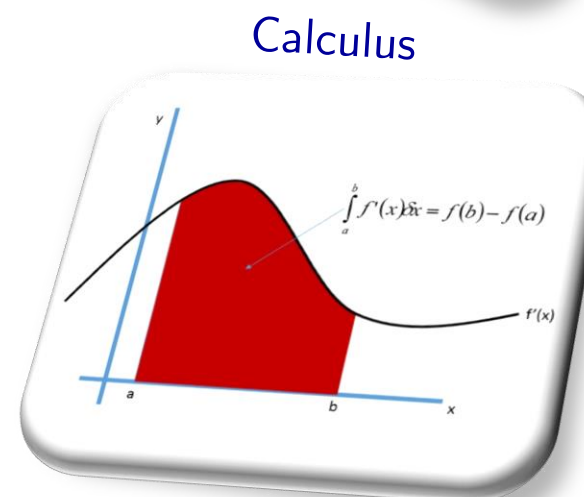
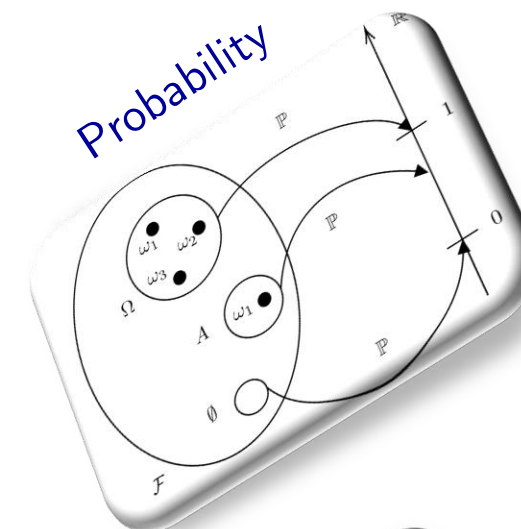
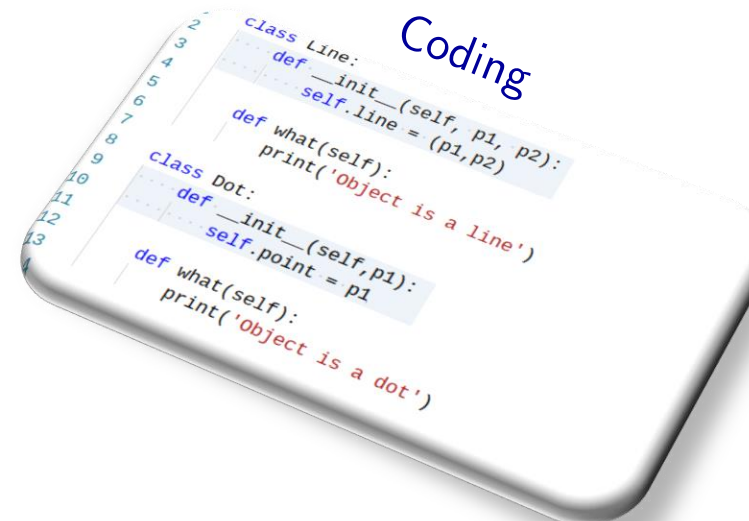
Objective of this Course

To develop a quantitative lens to look at finance and be empowered to apply quantitative ideas to the problems in finance easily, accurately and Independently



Valuation
Risk
Optimization

The Quant Symphony



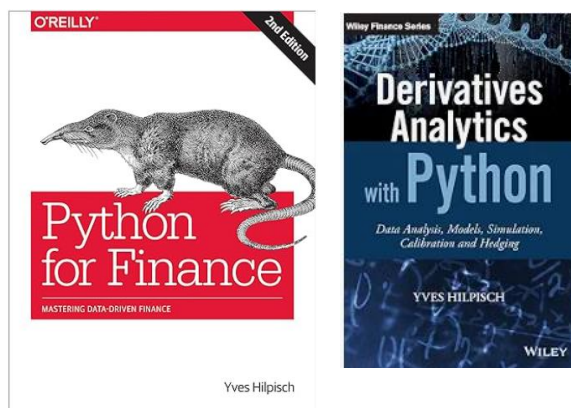
Managing Expectation



- ▶ Work hard with me (Attend, Practice problems, Experiment & Read)
- ▶ No spoon feeding
- ▶ Group decorum
- ▶ Flexibility (Important!)
- ▶ Faith

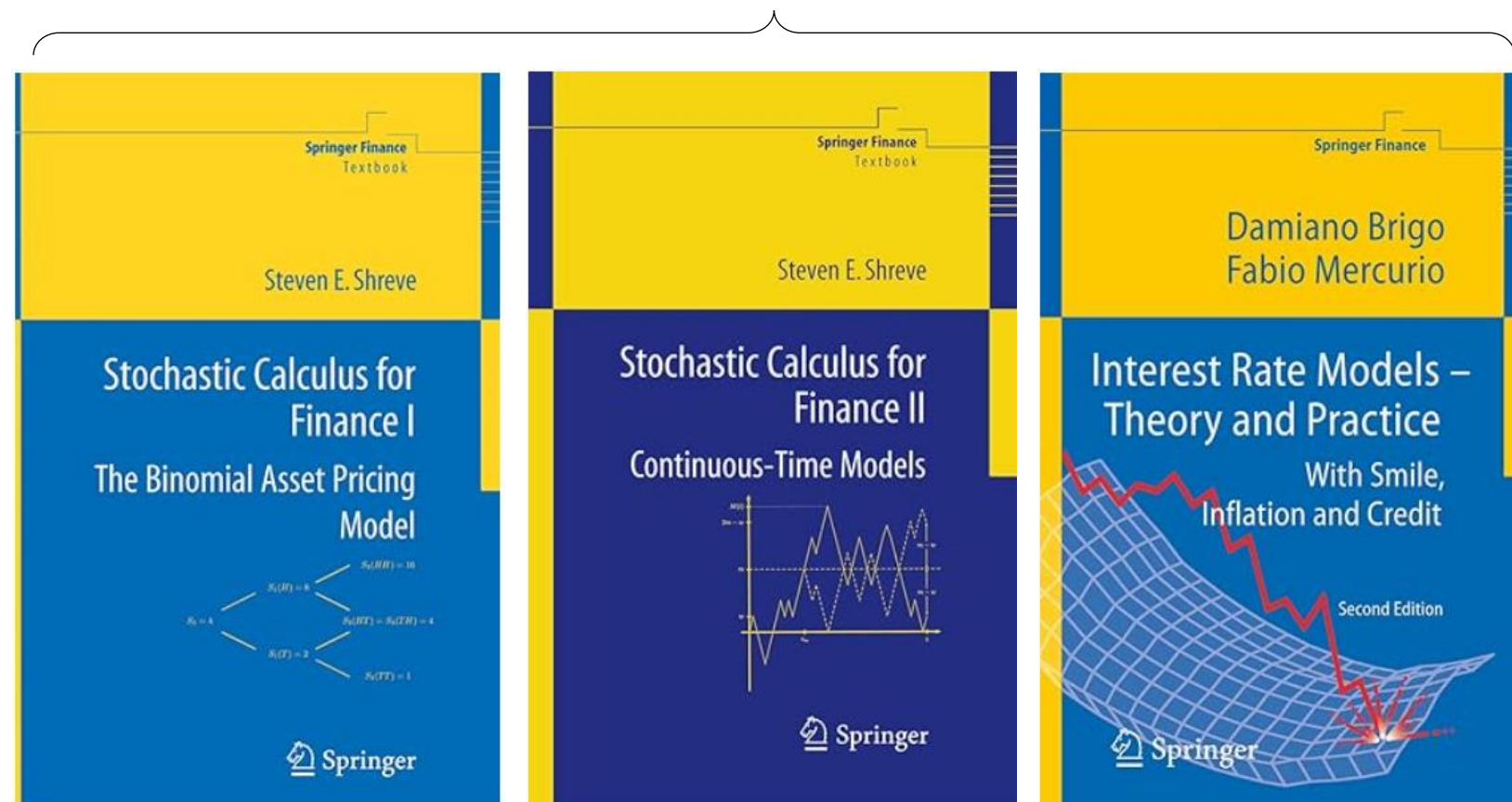
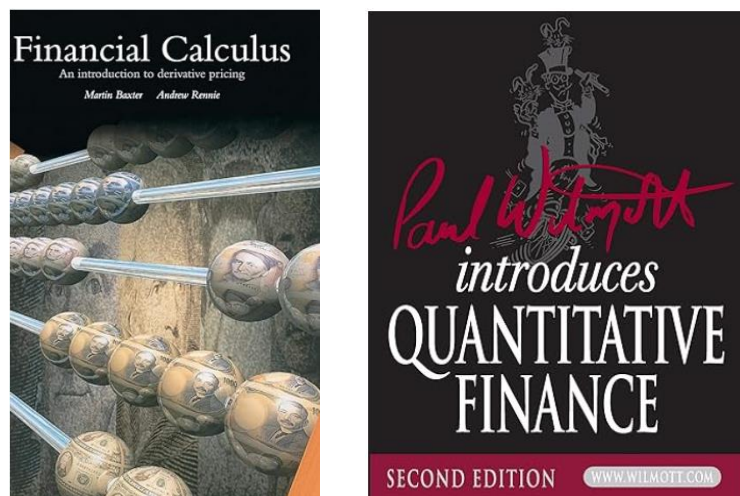
References

(Coding Focused)



(Rite of passage for a Quant)

(Easily digestible)



Class Timings & Resources

► Saturday , Sunday (5.30 pm IST to 7.30 pm IST) – (Live)

<https://peaks2tails.com/>

Class notes

Handwritten notes on ZCBs and interest rates:

$$Z(t, T) = \frac{P(t, T)}{P(t, t)} = P(t, T)$$

$$P(t, T) = e^{-\int_t^T R(s, T) ds}$$

$$\Rightarrow \langle P(t, T) = e^{-\int_t^T R(s, T) ds} \rangle$$

$$\langle R(t, T) = -\frac{1}{T-t} \ln \langle P(t, T) \rangle \rangle$$

xL and Python Models



Study Material

Lecture videos



Assignments



Valuation and Risk Modelling Python Assignment

Jupyter Notebook is to be emailed to peaks2tails@gmail.com

Equity

- Devise Monte Carlo pricing routine for and European Lookback call option with fixed strike. The payoff of the call option is $LC = \max(S_{max} - K, 0)$ where S_{max} is the maximum stock price over the life of the instrument. Assume continuous frequency (= daily simulation). Assume the following model parameters: $S_0 = \$100$, $r = 5\%$, $T = 1$ year, $K = \$100$, $\sigma = 20\%$. Answer the following questions in separate cells of the Jupyter Notebook.
 - Use explicit discretization of stock price on a daily frequency and Value of the Call option today. Compare with BSM price of a Vanilla European Call Option.
 - Find Monte Carlo Greeks: delta, gamma, vega, rho and theta. Hint: package Monte Carlo Pricing routine as a function and call the function multiple times with baseline

